

CMP-29 - Outcome and Success Evaluation Service

Evaluate Goal outcomes, side effects, constitutional costs, human impact, and declared success criteria using evidence and uncertainty.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 32. Authoritative state: Outcome Objects, success evaluations, side-effect records, review and learning referrals. Explicitly non-owned: Goals, Actions, Evidence Objects, Judgment, constitutional values.

Critical invariant

Success is evaluated from declared outcomes and constitutional costs, never from activity, completion, or self-reported confidence alone.

Logical interfaces

CMP-29-IF-01 / Command / Open Outcome Evaluation

Command `Open Outcome Evaluation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-02 / Command / Record Outcome

Command `Record Outcome` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-03 / Command / Evaluate Success

Command `Evaluate Success` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-04 / Command / Refer Learning

Command `Refer Learning` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-05 / Query / Get Outcome Object

Query `Get Outcome Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-06 / Query / Explain Success Result

Query `Explain Success Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-07 / Query / List Side Effects

Query `List Side Effects` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-08 / Event / Outcome Recorded

Event `Outcome Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-09 / Event / Success Evaluated

Event `Success Evaluated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-29-IF-10 / Event / Review Required

Event `Review Required` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Pending > Observing > Evaluated > Under Review > Closed or Reopened. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: vanity metric, missing side effect, premature evaluation, evidence cherry-pick, goal substitution. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-29-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-29-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-29-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-29-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-29-TST-005 - Critical invariant. Success is evaluated from declared outcomes and constitutional costs, never from activity, completion, or self-reported confidence alone. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-29-TST-006 - Failure containment. Inject vanity metric, missing side effect, premature evaluation, evidence cherry-pick, goal substitution. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-29-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-29-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-29-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-29-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.